

Rohit Kumar

Full Stack + AI/ML Developer

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PROFILE

Computer Science (Artificial Intelligence) student at Supaul College of Engineering, Bihar, building full-stack web applications and AI/ML solutions. Comfortable across React, Node.js, Express, MongoDB, REST APIs, NLP, and computer vision. Looking for internships, junior full-stack or AI/ML roles, and freelance work.

EDUCATION

B.Tech — Computer Science (Artificial Intelligence)

Supaul College of Engineering, Bihar · 2023 – 2027

Coursework and self-driven focus on full-stack web development, machine learning, NLP, and computer vision. Active member of the Technical Club.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C

Frontend: HTML, CSS, React, React Router, Vite

Backend: Node.js, Express.js, REST APIs, JWT, Role-Based Authorization

Databases: MongoDB, Mongoose, MongoDB Atlas

AI / ML: NLP, Computer Vision, Text Classification, TF-IDF, Logistic Regression, Naive Bayes, SVM, BERT, CNN/InceptionV3, ResNet, LSTM, MFCC Speaker Recognition

Tools: Git, GitHub, VS Code, Postman, Cloudinary, Razorpay, Nodemailer

PROJECTS

FaceID Attendance System

Tech Stack: Python, OpenCV, FaceNet, FastAPI, PostgreSQL, React

- Built an automated, contactless biometric attendance pipeline that detects and verifies faces from live video streams in under 500 ms.
- Integrated OpenCV Haar-cascade face bounding with FaceNet 128-dimensional embeddings, matched against an indexed PostgreSQL database using cosine similarity.
- Implemented blink and micro-motion detection heuristics to block photo spoofing, maintaining 98%+ verification accuracy.
- Developed a React reporting dashboard with automated daily CSV export and deployed the FastAPI backend on AWS EC2.

ResuMatch AI

Tech Stack: Next.js, TypeScript, Tailwind CSS, Prisma, PostgreSQL, OpenAI API

- Built a full-stack resume platform that scans target job descriptions and optimises resume structure, improving ATS match rates by ~45%.
- Integrated LLM endpoints with structured JSON prompts to parse keywords, score relevance, and suggest inline bullet-point improvements.
- Built an in-browser drag-and-drop editor using pdf-lib for client-side compilation, rendering downloadable PDFs in under 1 second.
- Deployed on Vercel with PostgreSQL managed through Prisma ORM and secure guest session saving.

AgroSense Solar IoT

Tech Stack: ESP32, C++, MQTT, Node.js, InfluxDB, React, Chart.js

- Designed an off-grid precision agriculture unit that automates irrigation from live soil telemetry, cutting simulated water use by ~35%.
- Programmed an ESP32 in C++ to sample capacitive soil moisture, temperature, and solar battery health using low-power deep-sleep cycles.
- Engineered an MQTT pipeline publishing telemetry every 30 seconds to a Node.js broker with fallback local relay control.
- Created a real-time portal visualising saturation graphs, thresholds, and pump history with React and Chart.js.

PulseCheck API Monitor

Tech Stack: Node.js, Express, Redis, BullMQ, PostgreSQL, Next.js, Resend API

- Built a distributed background service monitoring uptime, response latency, and SSL certificates across multiple endpoints.
- Implemented an asynchronous scheduler with Redis and BullMQ running non-blocking HTTP health checks at 1m/5m/15m intervals.
- Designed an indexed PostgreSQL schema tracking latency history, response payloads, and consecutive failures over 30-day windows.
- Integrated transactional email webhooks via the Resend API to dispatch down-alerts within 60 seconds of failure.

CodeFixer AI

Tech Stack: Next.js, TypeScript, Monaco Editor, FastAPI, LangChain, Redis, Tailwind CSS

- Engineered an in-browser code analysis environment that statically analyses code, flags runtime/logical defects, and proposes refactors.
- Embedded Monaco Editor connected to an asynchronous LangChain pipeline analysing time complexity, space complexity, and CWE vulnerabilities.
- Built a side-by-side diff viewer using WebStreams to stream optimised suggestions with inline additions and deletions.
- Implemented Redis hashing of snippets to serve cached reviews instantly, reducing LLM inference costs by ~60%.

Mini E-commerce Store

Tech Stack: React, Node.js, Express, MongoDB, Mongoose, JWT, Razorpay, Cloudinary, Nodemailer

- Built a MERN storefront with product catalogue, cart, and order flow backed by MongoDB and Mongoose schemas.
- Implemented JWT authentication with role-based authorization separating customer and admin capabilities.
- Integrated Razorpay test-mode payments, Cloudinary image uploads, and Nodemailer order confirmation emails.

Spam Email Classifier

Tech Stack: Python, NLTK, TF-IDF, Scikit-learn, Multinomial Naive Bayes

- Trained a TF-IDF + Multinomial Naive Bayes classifier on the 5,572-message SMS Spam Collection dataset.
- Achieved 97.3% accuracy with 96.1% precision, 89.4% recall, and a 92.6% F1-score.

Fake News Detector using NLP

Tech Stack: Python, Scikit-learn, TF-IDF, Logistic Regression, BERT, Flask, Streamlit

- Built a text-classification pipeline over ~44,000 labelled articles from the Kaggle Fake and Real News dataset.
- TF-IDF + Logistic Regression baseline reached 87% accuracy; the BERT version reached 91.4% accuracy (F1 91.4%).

ACHIEVEMENTS

- 7th and 6th rank holder in college assessments.
- Member of the college Technical Club.
- Participated in the Utkrisht college event, Webathon, a coding quiz, and a design portfolio activity.
- Best model accuracy achieved across ML projects: 97.3% (Spam Email Classifier).